Element H: Prototype Testing and Data Collection Plan

Student E, a student in our engineering class, needs a way to get rid of wasps, hornets, and other insects that enter his room at night when he sleeps. The student's bedroom has a serious insect problem. The insects make the space noisy and affect the students' ability to rest.

Our testing plan addresses the assembly time, size, weight, and effectiveness of the trap. We will use multiple series of tests in different locations for the most accurate data. The testing plan will cover the design criteria of an assembly time under 10 minutes, volume less than 14 cubic inches, weight less than one pound, and trapping the insects in one place.

The testing plan is as follows:

Assembly time under 10 minutes

1. Prepare materials to construct the design, as well as a stopwatch,
2. Start the stopwatch and assembly at the same time,
3. Stop the stopwatch when the assembly complete,
4. Record data,

Needs to be small

1. Measure assembled prototype,
2. Compare to size,
3. Record data,

Weight is less than one pound

1. Turn on the scale,
2. Tare scale,
3. Place prototype on scale,
4. Record data,

Efficiency

1. Peel off the insect paper cover,
2. Turn on the flashlight,
3. Place in the bedroom,
4. Leave for 8 hours from 10 pm to 6 am,
5. Record data,
6. Place outside,
7. Leave for 8 hours from 10 pm to 6 am,
8. Record data,
9. Place in the entryway,
10. Leave for 8 hours from 10 pm to 6 am,
11. Record data

The expert we chose to evaluate our testing procedures was Ms. Z. She was chosen due to her extensive experience overseeing student design projects and her strong understanding of the design process. Her feedback ensures that our testing approach is thorough and realistic. She reviewed our testing plan and made important improvements. She adjusted the order of the testing plan, which will prevent confusion when conducting the tests. She also encouraged us to specify criteria, such as the maximum volume of the prototype, and to conduct the tests in a variety of locations for the most accurate results.